Net force first

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Combine forces before using mass to predict acceleration.

SUCCESS CRITERIA - I can...

1. Find net force before dividing by mass.
2. State acceleration with units and direction.
3. Separate acceleration direction from motion direction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “net force” and “mass” to explain your model or result. Sentence starters are optional.

net force: the vector sum of all forces on one object

mass: a measure of how strongly an object resists acceleration

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

A 4 kg object hangs still from a vertical string.
Use g=9.8 N/kg.
Name all forces on it with sources, sizes and directions.
Does object-on-string belong on this diagram?

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: ΣF = ma; a = ΣF/m.
Add signed forces on the same object before dividing by mass. N/kg = m/s².

Worked reasoning: A 2 kg cart has 9 N right and 3 N left. Net force=6 N right; a=6/2=3 m/s² right. Add forces before dividing by mass.

Your turn: Draw arrows for both forces, then a separate net-force arrow.
Explain why 9/2 is not the acceleration.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



2 kg cart: draw the net force separately.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Work from force arrows

A 5 kg object has 32 N right and 12 N left. It currently moves left at 6 m/s.
Find a and v after 1 s under constant forces.

Sentence starter: I found ... [unit]. This value means ... in this situation.



The dot represents the object. Mass: 5 kg; velocity left

Marzano check - circle: ① ② ③

04 Find an unknown force

A 3 kg cart accelerates left at 2 m/s². One force is 4 N right; the other horizontal force is unknown. Complete the diagram and find the unknown force magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



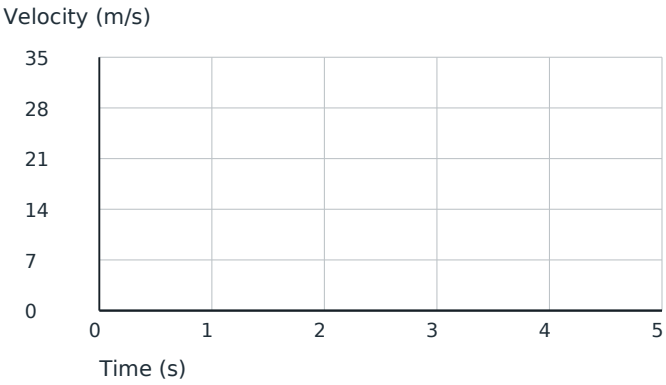
Mass 3 kg; acceleration 2 m/s² left.

Marzano check - circle: ① ② ③

05 From braking to force

A 1500 kg car brakes steadily from 30 m/s to rest in 5 s.
State a positive direction, plot the velocity-time line, then find acceleration and net force.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Positions are 21, 14, 7 m at 0, 2, 4 s.
Predict position at 9 s.
Give the rate, an assumption and a future test.
Separate what was observed from what can be inferred.

Sentence starter: I predict ... . My model supports this because ... .

Observation	Supported claim	Missing information
___	___	___
___	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 4 kg cart has 7 N right and 19 N left.
Find a. Is it necessarily moving left?
Name the information needed to determine whether it speeds up.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Add and label force arrows.

Draw the given forces; show net force separately.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 3 kg cart accelerates left at 2 m/s². A rope pulls right with 10 N.
Find the other horizontal force. Can the cart be moving right at this instant?

Sentence starter: I found ... [unit]. This value means ... in this situation.

Add and label force arrows.

Label each interaction force on the cart.

Marzano check - circle: 1 2 3